The empirical recurrence for OEIS A300092, recovered from its tail

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Abstract

OEIS A300092 records "Empirical recurrence of order 71" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{71} = -480 \neq 0$, so no further tail satisfies anything shorter; any order-71 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300092 is "Number of nX4 0..1 arrays with every element equal to 0, 2, 3, 5, 6, 7 or 8 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$1, 18, 19, 56, 203, 672, 2168, 7287, \ \ldots$$

The entry states:

> Empirical recurrence of order 71 (see link above)

As of the "Last modified" line on the live entry (Feb 25 2018 06:20:14, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 71: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 71.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 71 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{71} c_i \, a(n-i),$$

c_1	8	c_{19}	−19136	c_{37}	−9746509	c_{55}	78846749
c_2	−17	c_{20}	−98212	c_{38}	11775684	c_{56}	57293514
c_3	−3	c_{21}	−105001	c_{39}	33660770	c_{57}	11435899
c_4	16	c_{22}	30999	c_{40}	−51750490	c_{58}	1453262
c_5	9	c_{23}	−558853	c_{41}	−13873747	c_{59}	−10879948
c_6	152	c_{24}	195358	c_{42}	107068293	c_{60}	−4662487
c_7	−81	c_{25}	115529	c_{43}	105946840	c_{61}	−4184749
c_8	−702	c_{26}	1424147	c_{44}	37746540	c_{62}	4674248
c_9	775	c_{27}	1850822	c_{45}	−104011493	c_{63}	−1176901
c_{10}	412	c_{28}	−2451048	c_{46}	−1362957	c_{64}	−72650
c_{11}	−1654	c_{29}	−1181198	c_{47}	−29818455	c_{65}	164654
c_{12}	333	c_{30}	2398717	c_{48}	76904579	c_{66}	−196406
c_{13}	−8205	c_{31}	12798445	c_{49}	100465173	c_{67}	−102780
c_{14}	14936	c_{32}	5195419	c_{50}	−68852311	c_{68}	53196
c_{15}	11190	c_{33}	−8660705	c_{51}	−74087369	c_{69}	21856
c_{16}	22325	c_{34}	−7474833	c_{52}	−170230378	c_{70}	−4376
c_{17}	−66194	c_{35}	−27229088	c_{53}	−6872054	c_{71}	−480
c_{18}	68433	c_{36}	−29415100	c_{54}	−12371639		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A300092 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{71} - c_1 x^{71-1} - \cdots - c_{71}$. Its constant term is $-c_{71} = 480$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 71 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 71, so $\deg q = 71$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 71, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 71, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -480.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-71 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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